Computer Organization & Architecture

Unit- 1(Part 2)
Structure of Computers

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Performance

Term	Simple Definition	Example
Clock Rate (Clock Speed)	Number of clock cycles executed per second by the CPU	A **3 GHz** CPU performs **3 billion cycles per second**
CPI (Cycles Per Instruction)	Average number of clock cycles needed to execute one instruction	If one instruction takes **2 cycles on average**, CPI = **2**
Execution Time	Total time taken to run a program	A program taking **2 seconds** to finish
MIPS	Million instructions executed per second	If CPU executes **500 million instructions/sec**, MIPS = **500**

Complements

- Complements are used in digital computers **for simplifying the subtraction and logical manipulation.**
- Types of complements for each base r system:

r’s complements	**(r-1)’s complements**
10’s complement	9’s complement
2’s complement	1’s complement
8’s complement	7’s complement
16’s complement	15’s complement

(r-1)'s Complements

- If we are given a number N in base-r having n digits the (r-1)'s Complement is defined as:

(r^n - 1) - N

Example1:
Let us take N =1988. Here, r = 10 and n = 4, so 9's complement of 1988 is

9999 - 1988 = 8011

Example2: In binary we just change 1 from 0 and 0 from 1, we directly get its complement.
Let us take N = 010010 and r = 2 and n = 6. So directly 1's complement of N is

010010 = 1's complement is 101101, (Change all 0s to 1s and 1s to 0s).

r's Complements

- If we are given a number N in base-r having n digits the (r)'s Complement is defined as:

r^n - N

Example1:
Let N = 12345 and n = 5 and r = 10. So 10's complement of N is

100000 - 12345 = 87655

Example2: In binary, 2's complement is 1's complement +1 i.e. change 1 to 0 and 0 to 1 and then add 1 to the number.

The 2's complement of 0110111 is 1001001

Complements (Examples)

- Calculate 9's complement of $(2457)_{10}$
- Calculate 9’s complement of 134795
- Calculate 10's complement of $(5198)_{10}$
- Calculate 10’s complement of 134795
- Find the 1’s and 2’s complements of the binary number 1101001101
- Find the 1’s and 2’s complements of 100010100
- Explain how (r-1)’s complement is calculated. Calculate 9’s complement of 546700. (GTU)

Complements (Examples)

- Calculate 9's complement of $(2457)_{10}$ Ans:7542
- Calculate 9’s complement of 134795 Ans:865204
- Calculate 10's complement of $(5198)_{10}$ Ans:4802
- Calculate 10’s complement of 134795 Ans:865205
- Find the 1’s and 2’s complements of the binary number 1101001101

  Ans: 1’s complement: 0010110010 2’s complement: 0010110011
- Find the 1’s and 2’s complements of 100010100

  Ans: 1’s complement: 011101011 2’s complement: 011101100
- Explain how (r-1)’s complement is calculated. Calculate 9’s complement of 546700. (GTU) Ans: 453299

Subtraction using r's complement

To find M-N in base r, we add M + r's complement of N:

(More efficient than the method using paper and pencil in hardware)

1. Add the minuend M to the r's complement of the subtrahend N. This performs $M + (r^n - N) = M - N + r^n$.
2. If $M \geq N$, the sum will produce an end carry which is discarded, and what is left is the result $M - N$.
3. If $M < N$, the sum does not produce an end carry which is the r's complement of $(N - M)$. To obtain the answer in a familiar form, take the r's complement of the sum and place a negative sign in front.

Subtraction using r's complement (Examples)

For Decimal Numbers:

Consider, for example, the subtraction $72532 - 13250 = 59282$. The 10's complement of 13250 is 86750. Therefore:

$$
\begin{aligned}
M &= 72532 \\
\text{10's complement of } N &= +\underline{86750} \\
\text{Sum} &= 159282 \\
\text{Discard end carry } 10^5 &= -\underline{100000} \\
\text{Answer} &= 59282
\end{aligned}
$$

Now consider an example with $M < N$. The subtraction $13250 - 72532$ produces negative 59282. Using the procedure with complements, we have

$$
\begin{aligned}
M &= 13250 \\
\text{10's complement of } N &= +\underline{27468} \\
\text{Sum} &= 40718
\end{aligned}
$$

Here there is no end carry so, 10's complement of 40718 is 59282 and put a negative sign so that answer is (-59282)

Subtraction using r's complement (Examples)

For Binary Numbers: X=1010100 and Y=1000011

$$
\begin{array}{rr}
X = & 1010100 \\
2\text{'s complement of } Y = & +\underline{0111101} \\
\text{Sum} = & 10010001 \\
\text{Discard end carry } 2^7 = & -\underline{10000000} \\
\text{Answer: } X - Y = & 0010001
\end{array}
$$

$$
\begin{array}{rr}
Y = & 1000011 \\
2\text{'s complement of } X = & +\underline{0101100} \\
\text{Sum} = & 1101111
\end{array}
$$

There is no end carry

Answer is negative 0010001 = 2's complement of 1101111

Fixed point & Floating point representation

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Fixed point representation

- When an integer binary number is positive, the sign is represented by 0 and the magnitude by a positive binary number.
- When the number is negative, the sign is represented by 1 but the rest of the number may be represented :

    1. Signed-magnitude representation
    2. Signed-1's complement representation
    3. Signed 2's complement representation

- Although there is only one way to represent + 14(00001110), there are three different ways to represent - 14 with eight bits

    –In signed-magnitude representation 1 0001110

    –In signed-1's complement representation 1 11 10001

    –In signed-2's complement representation 1 11 10010

Arithmetic Addition

- Add the two numbers, including their sign bits, and discard any carry out of the sign (leftmost) bit position.
- Note that negative numbers must initially be in 2’s complement and that if the sum obtained after the addition is negative it is 2’s complement form.
- Adding numbers in the signed-2's complement system does not require a comparison or subtraction, only addition and complementation.

$$
\begin{array}{rl}
+6 & 00000110 \\
+13 & 00001101 \\
\hline
+19 & 00010011
\end{array}
\qquad
\begin{array}{rl}
-6 & 11111010 \\
+13 & 00001101 \\
\hline
+7 & 00000111
\end{array}
$$

$$
\begin{array}{rl}
+6 & 00000110 \\
-13 & 11110011 \\
\hline
-7 & 11111001
\end{array}
\qquad
\begin{array}{rl}
-6 & 11111010 \\
-13 & 11110011 \\
\hline
-19 & 11101101
\end{array}
$$

Arithmetic Subtraction

- Take the 2's complement of the subtrahend (including the sign bit) and add it to the minuend (including the sign bit). A carry out of the sign bit position is discarded.
- We could say that the subtraction operation can be changed to an addition operation if the sign of the subtrahend is changed. This is demonstrated by the following relationship:

- Here, 5-3=2
- 5+(-3)=2

$$(\pm A) - (+B) = (\pm A) + (-B)$$

- (-8)-(-3)=(-5)
- (-8)+(3)=(-5)

$$(\pm A) - (-B) = (\pm A) + (+B)$$

- Perform A – B (subtract) operation for the following numbers using signed magnitude number format. A = + 11 and B = - 6.
- A = **+11**

  B = **−6**
- We have to perform:

  A−B=+11−(−6)=+11+6

  1011+0110=10001
- The result of the addition is 10001

  Sign bit: 0(positive)

  Magnitude: 10001

Floating-Point Representation

- Used to represent very small and very large numbers.
- Two parts
  - Mantissa – Represents signed fixed-point number
  - Exponent – Represents position of the decimal point
- For example, the decimal number +6132.789 is represented in floating-point with a fraction and an exponent as follows:

Fraction	Exponent
+0.6132789	+04

- The value of the exponent indicates that the actual position of the decimal point is four positions to the right of the indicated decimal point in the fraction.
- Floating-point is always interpreted to represent a number in the following form: $m \times r^e$
- Here, m stands for mantissa, e stands for exponent, r means radix
- Scientific Equivalent : $+0.6132789 \times 10^{+4}$

Floating-Point Representation

- Normalization
  - A floating point number is said to be normalized if the most significant digit of the mantissa is nonzero.
  - For example, the decimal number 350 is normalized but 00035 is not.
- ANSI 32-bit floating point byte format

Byte 1	Byte 2	Byte 3	Byte 4
SEEEEEEE	.MMMMMMMM	MMMMMMMM	MMMMMMMM

- *S = Sign of Mantissa, E = Exponent bits, M = Mantissa bits*

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Floating-Point Representation

- ANSI 32-bit floating point byte format

Byte 1	Byte 2	Byte 3	Byte 4
SEEEEEEE	.MMMMMMMM	MMMMMMMM	MMMMMMMM

- *S = Sign of Mantissa, E = Exponent bits, M = Mantissa bits*

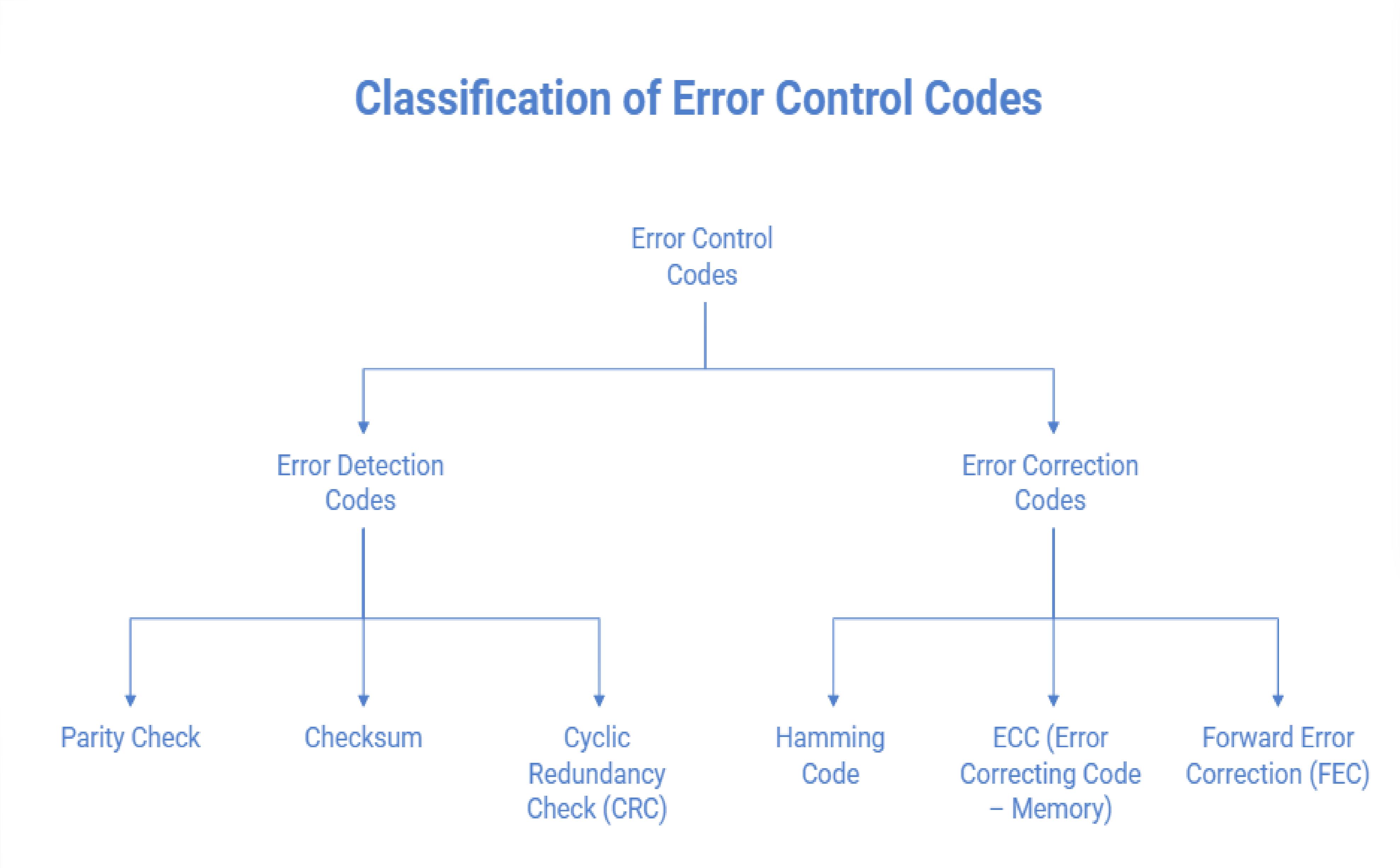
- Example, 13 = 1101 = 0.1101 x 2^4

  = 00000100 .11010000 00000000 00000000

- Example, -17 = - 10001 = -0.10001 x 2^5

  = 10000101 .10001000 00000000 00000000

Error Detection and Correction Codes



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Error Detection

1. Error Detection Codes

- Error Detection Codes are used to **detect whether an error has occurred** during data transmission.
  They **cannot correct** the error; they only indicate that data is incorrect.

(a) Parity Check

- An extra bit called **parity bit** is added to the data.
- It makes the total number of 1s either **even** (even parity) or **odd** (odd parity).
- **Example:**
  Data = 1011001 (number of 1s = 4 → even)
  Even parity bit = 0
  Transmitted data = 10110010
- If one-bit changes during transmission, parity becomes incorrect → **error detected**.
- **Limitation:**
- Detects only **single-bit errors**

Error Detection

(b) Checksum

- Data is divided into blocks.
- All blocks are added together to form a checksum.
- The checksum is sent along with the data.
- Receiver recalculates and compares the checksum.
- **Example:**
  Data blocks: 1001, 0011
  Checksum = 1001 + 0011 = 1100
  If the received checksum does not match → error detected.
- Used in:
- File transfer
- Network communication

Error Detection

(c) Cyclic Redundancy Check (CRC)

- Data is treated as a **binary number**.
- It is divided by a **generator polynomial**.
- The remainder is appended to the data.
- **Example (conceptual):**
  Data = 110101
  Generator = 1011
  Remainder = 100
  Transmitted data = 110101100
- CRC is very powerful and can detect **burst errors**.

Used in:

- Ethernet
- Hard disks
- Communication protocols

Error Correction Codes

- Error Correction Codes can **detect and correct errors automatically** without retransmission.

(a) Hamming Code

- Extra parity bits are added at specific positions (1, 2, 4, 8…).
- These bits help locate the **exact position of the error**.

(b) ECC (Error Correcting Code – Memory)

- Special type of Hamming-based code used in **RAM**.
- Automatically corrects memory bit errors.

(c) Forward Error Correction (FEC)

- Redundant bits are sent **in advance**.
- Receiver corrects errors without asking for retransmission.

GTU Questions

Q.1 Explain signed representation of integer in computer. Write (–12) in binary with 8 – bits using following representations (i) Signed magnitude (ii) Signed 1’s complement (iii) Signed 2’s complement.

- Only one way to represent + 12(00001100), there are three different ways to represent (-12) with eight bits
  - –In signed-magnitude representation 1 0001100
  - –In signed-1's complement representation 1 1110011
  - –In signed-2's complement representation 1 1110100

GTU Questions

Q.2 Perform A – B (subtract) operation for the following numbers using signed magnitude number format. (Write necessary assumptions if required) A = + 11 and B = - 6.

- Assume we are using 5 bits (1 sign bit + 4 magnitude bits) to represent the numbers:

  Sign bit: 0 for positive and 1 for negative

  Magnitude bits: Represent the absolute value of the number.

- A=+11 (0 1011)   B=-6 (1 0110)
- Now, Add the magnitudes of A and B, keeping the sign bit separate.

  A−B=+11−(−6)=+11+6

  1011+0110=10001

- The result of the addition is 10001

  Sign bit: 0(positive)

  Magnitude: 10001

GTU Questions

Q.3 What is binary and decimal equivalent of F8 hexadecimal value?

1. Hexadecimal to Binary Conversion

   Each hexadecimal digit represents 4 bits in binary:

   F=1111(binary for 15 in decimal)

   8=1000(binary for 8 in decimal)

   So, the binary equivalent of F8is:

   $\mathbf{(F8)_{16}=(1111\ 1000)_2}$

2. Hexadecimal to Decimal Conversion

   To convert $(F8)_{16}$ to decimal:

   $F8_{16}=(15\times16^1)+(8\times16^0)$

   $F8_{16}=(15\times16)+(8\times1)$

   $\mathbf{(F8)_{16}=240+8=(248)_{10}}$

GTU Questions

Q.4 Explain how (r-1)’s complement is calculated. Calculate 9’s complement of 546700.

Here r=10, n=6 and N= 546700

9’s complement of 546700 is calculated as:

=(r^n - 1) – N

=(106-1)-(546700)

=(1000000-1)-(546700)

=999999- 546700

= 453299


Thank you!